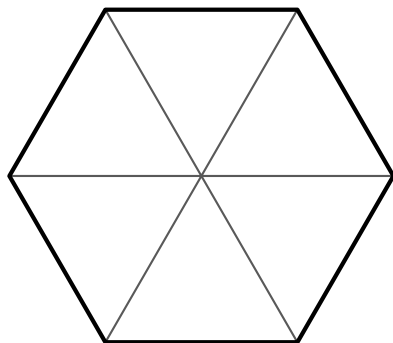
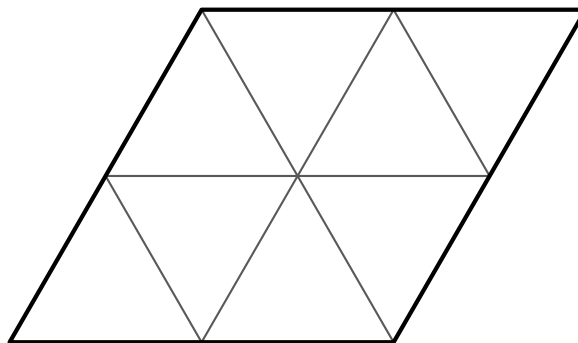


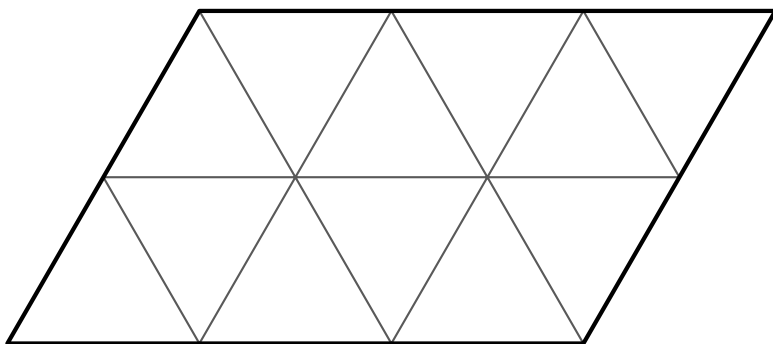
Problem 1: Play this game with a partner. Take turns placing one blue rhombus on the board so that it covers two small triangles and does not overlap another rhombus. The player who cannot place a rhombus loses. Play several games on each board below, and circle the player who can always win there.



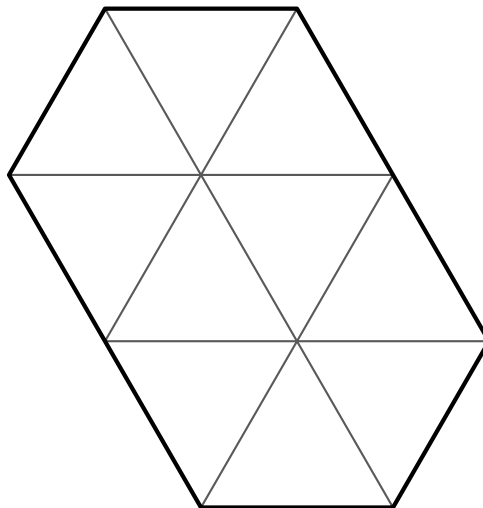
1st 2nd



1st 2nd

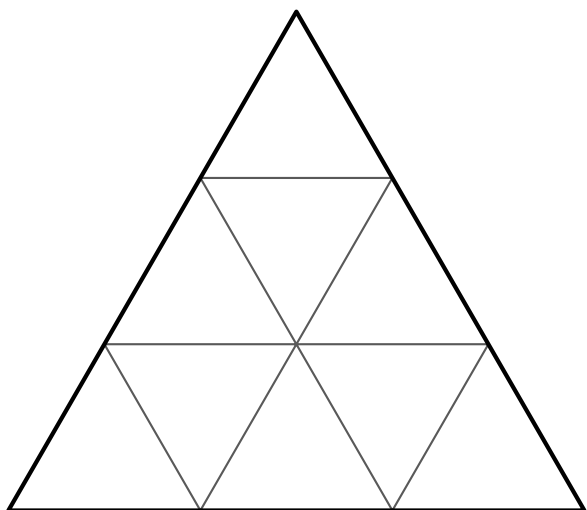


1st 2nd

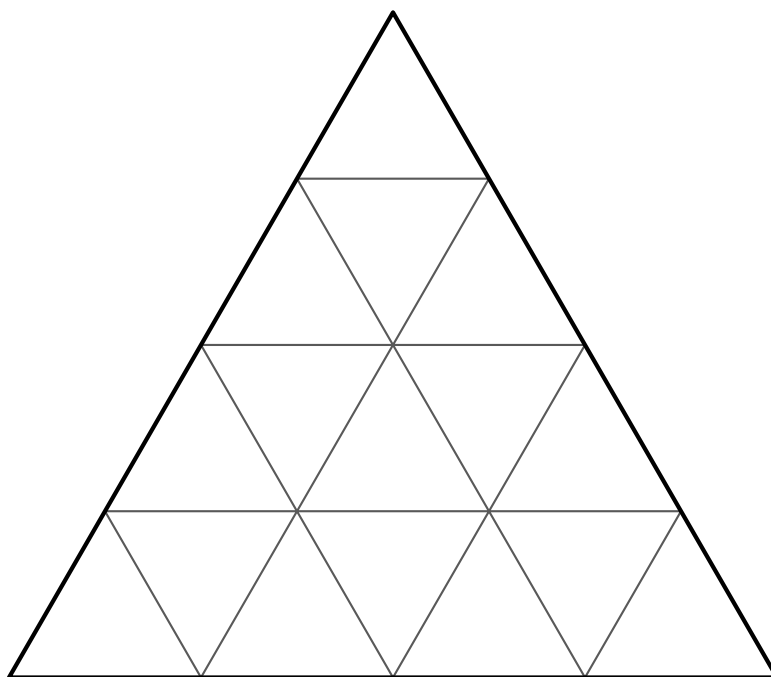


1st 2nd

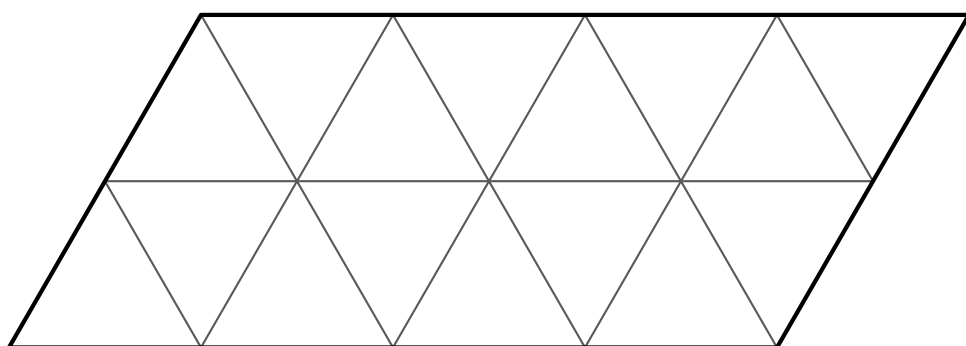
Problem 2: Play the game from Problem 1 on these three boards. Circle the player who can always win on each one.



1st 2nd

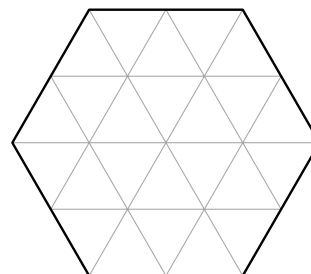
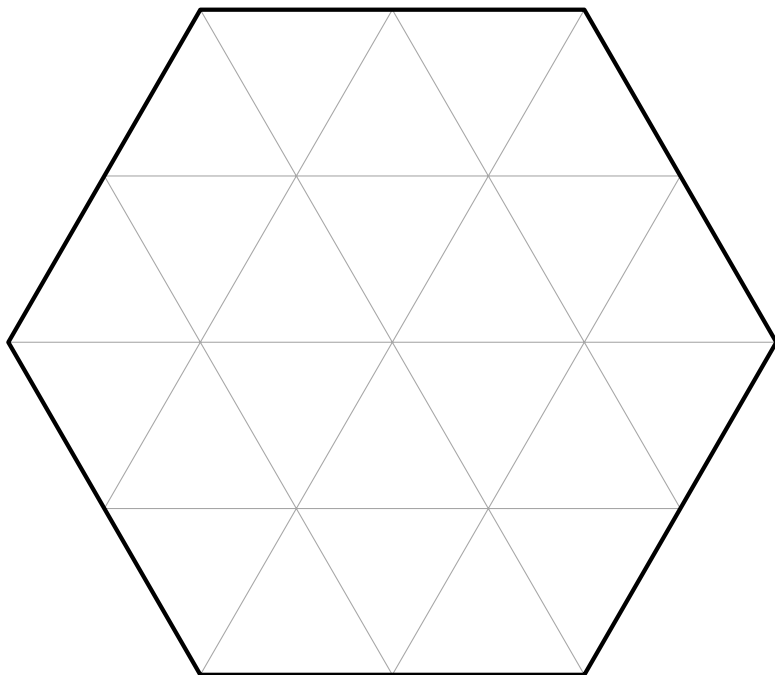


1st 2nd

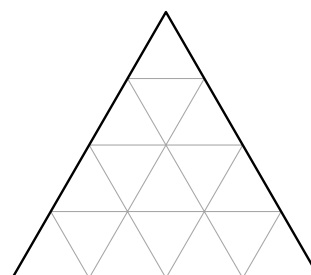
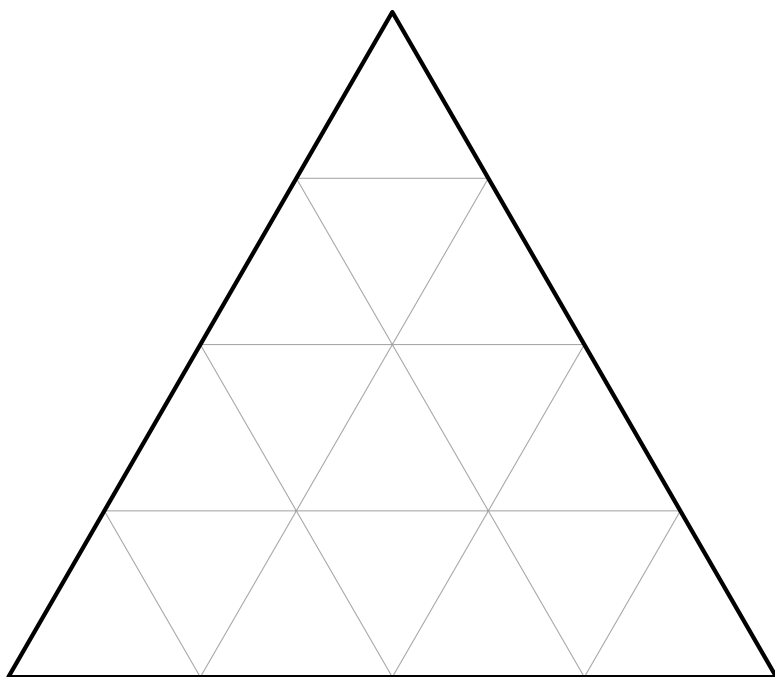


1st 2nd

Problem 3: Fill each board with as few pieces as you can, using only the small green triangles, blue rhombi, red trapezoids and yellow hexagons. Draw your best filling on the small copy and write how many pieces it has. Explain why the hexagon cannot be filled with fewer pieces.

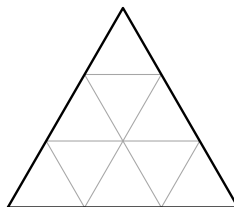
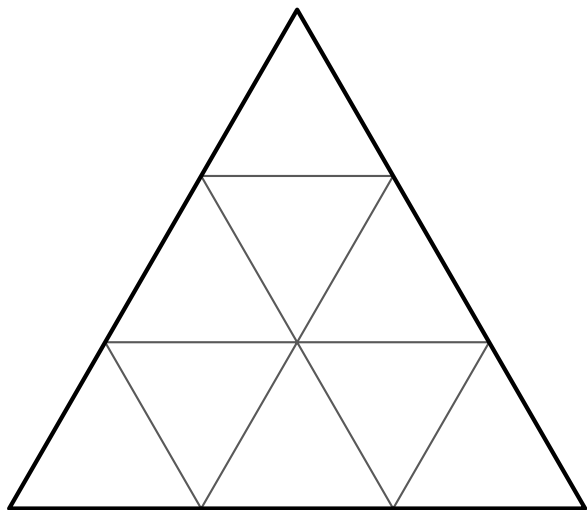


pieces

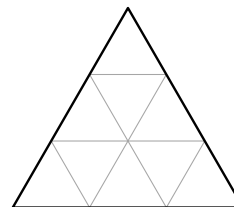


pieces

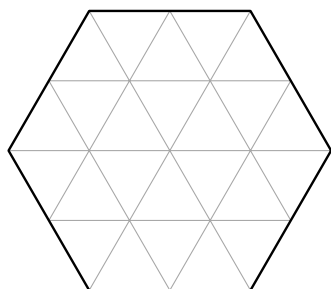
Problem 4: Fill the triangle below with small red trapezoids in every way you can, and fill the hexagon from Problem 3 with small red trapezoids in several different ways. Draw each filling on a small copy. Write how many of its trapezoids cover two up-triangles and how many cover two down-triangles.



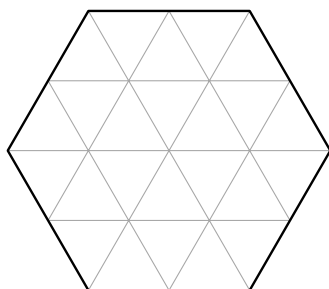
two up: _____
two down: _____



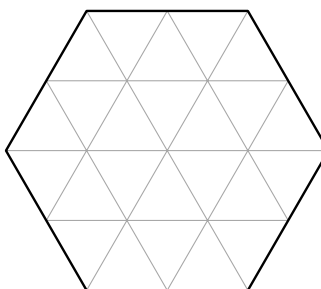
two up: _____
two down: _____



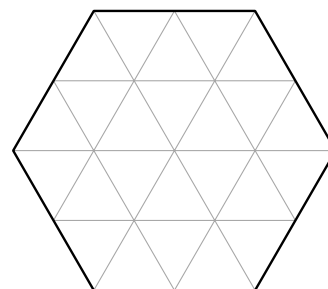
two up: _____
two down: _____



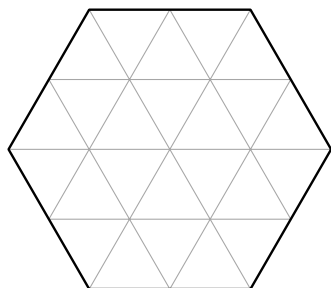
two up: _____
two down: _____



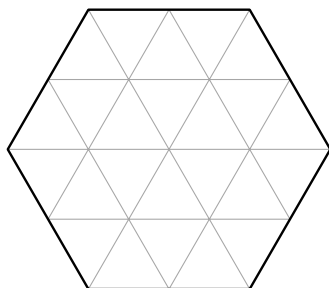
two up: _____
two down: _____



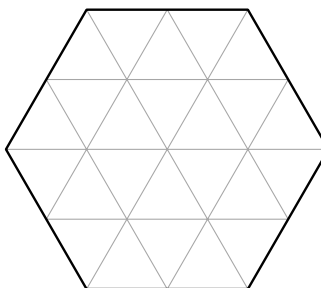
two up: _____
two down: _____



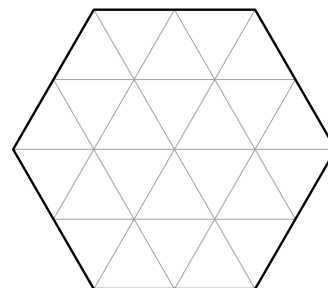
two up: _____
two down: _____



two up: _____
two down: _____

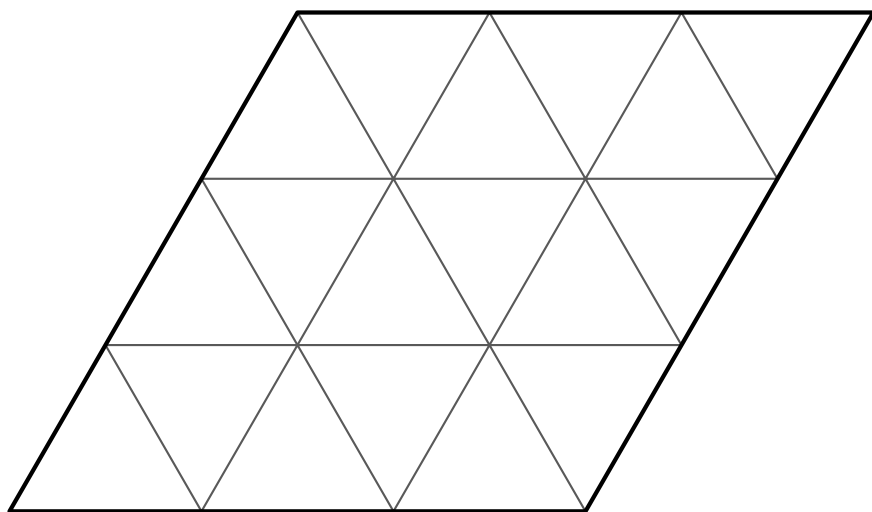
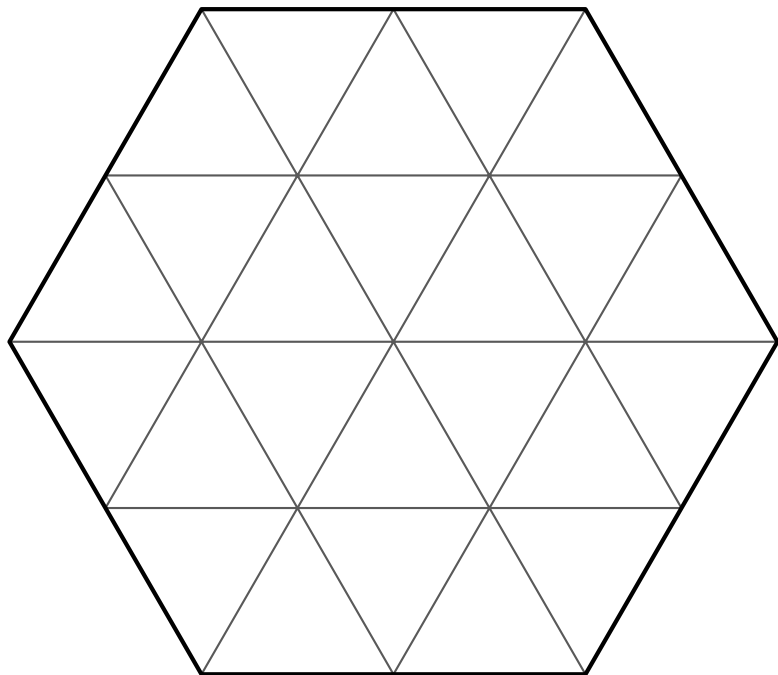


two up: _____
two down: _____

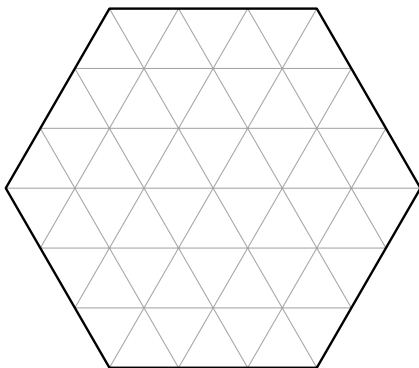
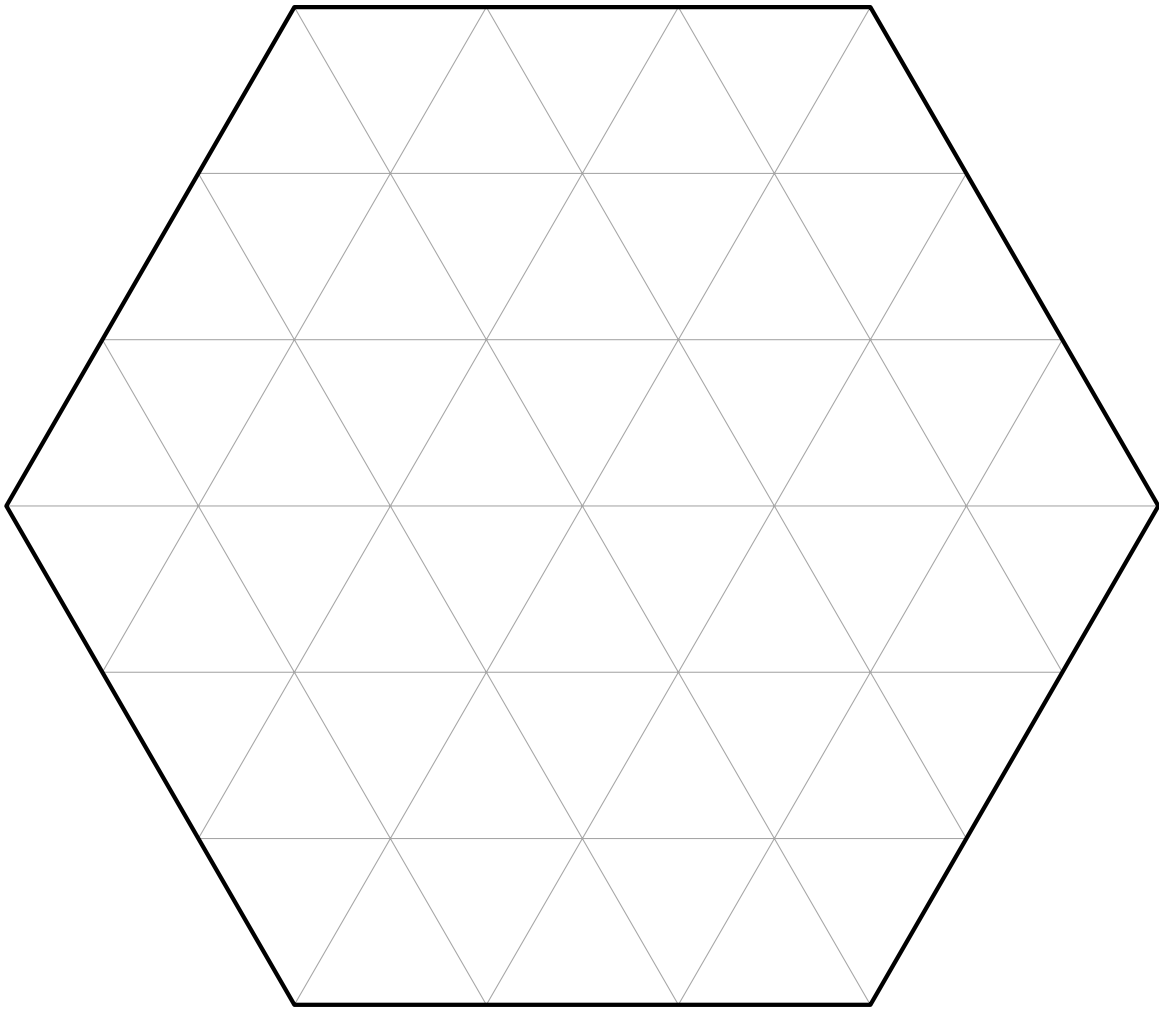


two up: _____
two down: _____

Problem 5: Play the game on these two boards. On each board, find a way for one of the players to win every game, and explain why it always works.



Problem 6: Fill this big hexagon with as few pieces as you can, using only the small green triangles, blue rhombi, red trapezoids and yellow hexagons. Draw your best filling on the small copy and write how many pieces it has.



pieces

Problem 7: Fill this big triangle with small red trapezoids. Try to make as many of them as you can cover two down-triangles. What is the most you can get? Explain why you cannot get more.

